U.S. FOOD & DRUG ADMINISTRATION

June 16, 2025

Siemens Healthcare GmbH
Alina Goodman
Regulatory Affairs Professional
Siemens Medical Solutions USA, Inc.
40 Liberty Boulevard
Malvern, Pennsylvania 19355

Re: K250443
Trade/Device Name: MAGNETOM Avanto Fit;
MAGNETOM Skyra Fit;
MAGNETOM Sola Fit;
MAGNETOM Viato.Mobile
Regulation Number: 21 CFR 892.1000
Regulation Name: Magnetic Resonance Diagnostic Device
Regulatory Class: Class II
Product Code: LNH, LNI, MOS
Dated: May 16, 2025
Received: May 16, 2025

Dear Alina Goodman:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (https://www.fda.gov/media/99812/download) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (https://www.fda.gov/media/99785/download).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance) and CDRH Learn (https://www.fda.gov/training-and-continuing-education/cdrh-learn). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Daniel M. Krainak, Ph.D.
Assistant Director
DHT8C: Division of Radiological
Imaging and Radiation Therapy Devices
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

Please type in the marketing application/submission number, if it is known. This textbox will be left blank for original applications/submissions.	K250433	?
Please provide the device trade name(s).		?

MAGNETOM Avanto Fit;
MAGNETOM Skyra Fit;
MAGNETOM Sola Fit;
MAGNETOM Viato.Mobile

Please provide your Indications for Use below. ?

The MAGNETOM system is indicated for use as a magnetic resonance diagnostic device (MRDD) that produces transverse, sagittal, coronal and oblique cross sectional images, spectroscopic images and/or spectra, and that displays the internal structure and/or function of the head, body, or extremities. Other physical parameters derived from the images and/or spectra may also be produced. Depending on the region of interest, contrast agents may be used. These images and/or spectra and the physical parameters derived from the images and/or spectra when interpreted by a trained physician yield information that may assist in diagnosis.

The MAGNETOM system may also be used for imaging during interventional procedures when performed with MR compatible devices such as in-room displays and MR Safe biopsy needles.

Please select the types of uses (select one or both, as applicable).	☒ Prescription Use (Part 21 CFR 801 Subpart D) ☐ Over-The-Counter Use (21 CFR 801 Subpart C)	?

K250443

510(k) Summary

This summary of 510(k) safety and effectiveness information is being submitted in accordance with the requirements of the Safe Medical Devices Act 1990 and 21 CFR § 807.92.

1. General Information

Establishment:
Siemens Medical Solutions USA, Inc.
40 Liberty Boulevard
Malvern, PA 19355, USA
Registration Number: 2240869

Date Prepared:
February 3rd, 2025

Manufacturer:
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Registration Number: 3002808157

Siemens Shenzhen Magnetic Resonance LTD.
Siemens MRI Center
Hi-Tech Industrial Park (middle)
Gaoxin C. Ave., 2nd
Shenzhen 518057
P.R. CHINA
Registration Number: 3004754211

2. Contact Information

Alina Goodman
Regulatory Affairs Professional
Siemens Medical Solutions USA, Inc.
40 Liberty Boulevard
Malvern, PA 19355, USA
Phone: +1(317)371-8593
E-mail: alina.goodman@siemens-healthineers.com

3. Device Name and Classification

Device/ Trade name:	MAGNETOM Avanto Fit
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Device/ Trade name:	MAGNETOM Skyra Fit
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Device/ Trade name:	MAGNETOM Sola Fit
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Device/ Trade name:	MAGNETOM Viato.Mobile
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Device/ Trade name:

4. Legally Marketed Predicate and Reference Device

4.1. Predicate Device

Trade name:	MAGNETOM Avanto Fit
510(k) Number:	K220151
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Trade name:	MAGNETOM Skyra Fit
510(k) Number:	K220589
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Trade name: MAGNETOM Sola Fit[1]

[1] The predicate system MAGNETOM Sola Fit (1.5T) with syngo MR XA51A (K221733) is used in this submission as well as reference system in the Fact Sheets for some systems. However, it will not be listed again as reference system since it is already incorporated into the predicate system overview.

510(k) Number: K221733
Classification Name: Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel: Radiology
CFR Code: 21 CFR § 892.1000
Classification: II
Product Code: Primary: LNH
Secondary: LNI, MOS

Trade name: MAGNETOM Viato.Mobile
510(k) Number: K240608
Classification Name: Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel: Radiology
CFR Code: 21 CFR § 892.1000
Classification: II
Product Code: Primary: LNH
Secondary: LNI, MOS

Trade name: syngo.via VB40A
510(k) Number: K191040
Classification Name:
Classification Panel:
CFR Code:
Classification:
Product Code: Primary: LLZ

4.2. Reference Device

Trade name: MAGNETOM Cima.X
510(k) Number: K231587
Classification Name: Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel: Radiology
CFR Code: 21 CFR § 892.1000
Classification: II
Product Code: Primary: LNH
Secondary: LNI, MOS

Trade name: MAGNETOM Sola
510(k) Number: K232535
Classification Name: Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel: Radiology
CFR Code: 21 CFR § 892.1000
Classification: II
Product Code: Primary: LNH
Secondary: LNI, MOS

Trade name: MAGNETOM Vida
510(k) Number: K213693
Classification Name: Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel: Radiology

CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

Trade name:	MAGNETOM Aera
510(k) Number:	K153343
Classification Name:	Magnetic Resonance Diagnostic Device (MRDD)
Classification Panel:	Radiology
CFR Code:	21 CFR § 892.1000
Classification:	II
Product Code:	Primary: LNH Secondary: LNI, MOS

4. Intended Use / Indications for Use

The indications for use for the subject devices are the same as the predicate devices:

The MAGNETOM system is indicated for use as a magnetic resonance diagnostic device (MRDD) that produces transverse, sagittal, coronal and oblique cross sectional images, spectroscopic images and/or spectra, and that displays the internal structure and/or function of the head, body, or extremities. Other physical parameters derived from the images and/or spectra may also be produced. Depending on the region of interest, contrast agents may be used. These images and/or spectra and the physical parameters derived from the images and/or spectra when interpreted by a trained physician yield information that may assist in diagnosis.

The MAGNETOM system may also be used for imaging during interventional procedures when performed with MR compatible devices such as in-room displays and MR Safe biopsy needles.

5. Device Description

The subject device, MAGNETOM Avanto Fit with software syngo MR XA70A, consists of new and modified software and hardware that is similar to what is currently offered on the predicate device, MAGNETOM Avanto Fit with syngo MR XA50A (K220151).

A high-level summary of the new and modified hardware and software is provided below:

For MAGNETOM Avanto Fit with syngo MR XA70:
Hardware
New Hardware:
- ➔ **myExam 3D Camera**
- ➔ **BM Head/Neck 20**

Modified Hardware:
- ➔ **-Sanaflex (cushions for patient positioning)**

Software
New Features and Applications:
- ➔ **myExam Autopilot Brain**
- ➔ **myExam Autopilot Knee**
- ➔ **3D Whole Heart**
- ➔ **HASTE_interactive**
- ➔ **GRE_PC**
- ➔ **Open Recon**
- ➔ **Deep Resolve Gain**
- ➔ **Fleet Reference Scan**
- ➔ **Physio logging**
- ➔ **complex averaging**
- ➔ **AutoMate Cardiac**
- ➔ **Ghost Reduction**
- ➔ **BLADE diffusion**
- ➔ **Beat Sensor**
- ➔ **Deep Resolve Sharp**
- ➔ **Deep Resolve Boost and Deep Resolve Boost (TSE)**
- ➔ **Deep Resolve Boost HASTE**
- ➔ **Deep Resolve Boost EPI Diffusion**

Modified Features and Applications:
- ➔ **SPACE improvement (high band)**
- ➔ **SPACE improvement (incr grad)**
- ➔ **Brain Assist**
- ➔ **Eco power mode**
- ➔ **myExam Angio Advanced Assist (Test Bolus)**

The subject device, MAGNETOM Skyra Fit with software syngo MR XA70A, consists of new and modified software and hardware that is similar to what is currently offered on the predicate device, MAGNETOM Skyra Fit with syngo MR XA50A (K220589).

A high-level summary of the new and modified hardware and software is provided below:

<u>For MAGNETOM Skyra Fit with syngo MR XA70:</u>
<u>Hardware</u>
<u>New Hardware:</u>
- ➔ **myExam 3D Camera**

<u>Modified Hardware:</u>
- ➔ **Sanaflex (cushions for patient positioning)**

<u>Software</u>
<u>New Features and Applications:</u>
- ➔ **Beat Sensor**
- ➔ **HASTE_interactive**
- ➔ **GRE_PC**
- ➔ **3D Whole Heart**
- ➔ **Deep Resolve Gain**
- ➔ **Open Recon**
- ➔ **Ghost Reduction**
- ➔ **Fleet Reference Scan**
- ➔ **BLADE diffusion**
- ➔ **HASTE diffusion**
- ➔ **Physio logging**
- ➔ **complex averaging**
- ➔ **Deep Resolve Swift Brain**
- ➔ **Deep Resolve Sharp**
- ➔ **Deep Resolve Boost and Deep Resolve Boost (TSE)**
- ➔ **Deep Resolve Boost HASTE**
- ➔ **Deep Resolve Boost EPI Diffusion**
- ➔ **AutoMate Cardiac**
- ➔ **SVS_EDIT**

<u>Modified Features and Applications:</u>
- ➔ **SPACE improvement (high band)**
- ➔ **SPACE improvement (incr grad)**
- ➔ **Brain Assist**
- ➔ **Eco power mode**
- ➔ **myExam Angio Advanced Assist (Test Bolus)**

The subject device, MAGNETOM Sola Fit with software syngo MR XA70A, consists of new and modified software and hardware that is similar to what is currently offered on the predicate device, MAGNETOM Sola Fit with syngo MR XA51A (K221733).

A high-level summary of the new and modified hardware and software is provided below:

<u>For MAGNETOM Sola Fit with syngo MR XA70:</u>
<u>Hardware</u>
<u>New Hardware:</u>
- ➔ **myExam 3D Camera**

<u>Modified Hardware:</u>
- ➔ **Sanaflex (cushions for patient positioning)**

<u>Software</u>
<u>New Features and Applications:</u>

- **GRE_PC**
- **3D Whole Heart**
- **Ghost Reduction**
- **Fleet Reference Scan**
- **BLADE diffusion**
- **Physio logging**
- **Open Recon**
- **Complex averaging**
- **Deep Resolve Sharp**
- **Deep Resolve Boost and Deep Resolve Boost (TSE)**
- **Deep Resolve Boost HASTE**
- **Deep Resolve Boost EPI Diffusion**
- **AutoMate Cardiac**
- **Implant suite**

Modified Features and Applications:

- **SPACE improvement (high band)**
- **SPACE improvement (incr grad)**
- **Brain Assist**
- **Eco power mode**

The subject device, MAGNETOM Viato.Mobile with software syngo MR XA70A, consists of new and modified software and hardware that is similar to what is currently offered on the predicate device, MAGNETOM Viato.Mobile with syngo MR XA51A (K240608).

A high-level summary of the new and modified hardware and software is provided below:

For MAGNETOM Viato.Mobile with syngo MR XA70:

Hardware

New Hardware:

- **n.a.**

Modified Hardware:

- **Sanaflex (cushions for patient positioning)**

Software

New Features and Applications:

- **GRE_PC**
- **3D Whole Heart**
- **Ghost Reduction**
- **Fleet Reference Scan**
- **BLADE diffusion**
- **Physio logging**
- **Open Recon**
- **Complex averaging**
- **Deep Resolve Sharp**
- **Deep Resolve Boost and Deep Resolve Boost (TSE)**
- **Deep Resolve Boost HASTE**
- **Deep Resolve Boost EPI Diffusion**
- **AutoMate Cardiac**
- **Implant suite**

Modified Features and Applications:

- **SPACE improvement (high band)**
- **SPACE improvement (incr grad)**

➔ **Brain Assist**
➔ **Eco power mode**

Furthermore, the following minor updates and changes were conducted for the subject devices:

- Low SAR Protocol minor update (for all subject devices but MAGNETOM Skyra Fit): the goal of the SAR adaptive protocols was to be able to perform knee, spine, heart and brain examinations with 50% of the max allowed SAR values in normal mode for head and whole-body SAR. The SAR reduction was achieved by parameter adaptations like Flip angle, TR, RF Pulse Type, Turbo Factor, concatenations. For cardiac clinically accepted alternative imaging contrasts are used (submitted with K232494).
- Implementation of image sorting prepare for PACS (submitted with K231560).
- Implementation of improved DICOM color support (submitted with K232494).
- Needle intervention AddIn was added all subject device (submitted with K232494).
- Inline Image Filter switchable for users: in the subject device, users have the ability to switch the “Inline image filter” (implicite Filter) on or off. This filter is an image-based filter that can be applied to specific pulse sequence types. The function of the filter remains unchanged from the previous device MAGNETOM Sola with syngo MR XA61A (K232535).
- SVS_EDIT is newly added for MAGNETOM Skyra Fit, but without any changes (submitted with K203443)
- Brain Assist received an improvement and is identical to that of snygo MR XA61A (K232535)
- Open Recon is introduced for all systems. The function of Open Recon remains unchanged from the previous submissions (submitted with K221733).
- Lock TR and FA in Bold received a minor UI update
- Implant Suite is newly introduced for MAGNETOM Sola Fit and MAGNETOM Viato.Mobile, but without any changes (submitted with K232535)
- myExam Autopilot Brain and myExam Autopilot Knee are newly introduced for the subject device MAGNETOM AVANTO Fit and are unchanged from previous submissions (submitted with K221733).
- myExam Angio Advanced Assist (Test Bolus) received a bug fixing and minimal UI improvements

6. Substantial Equivalence

MAGNETOM Avanto Fit, MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile, all with software syngo MR XA70A are substantially equivalent to the following devices:

Predicate Device	FDA Clearance Number and Date	Product Code	Manufacturer
MAGNETOM Avanto Fit (1.5T) with syngo MR XA50A	K220151, cleared on April 01, 2022	LNH, LNI, MOS	Siemens Healthcare GmbH
MAGNETOM Skyra Fit (3T) with syngo MR XA50A	K220589, cleared on May 13, 2022	LNH, LNI, MOS	Siemens Healthcare GmbH

MAGNETOM Sola Fit[2] (1.5T) with syngo MR XA51A	K221733, cleared on September 13, 2022	LNH, LNI, MOS	Siemens Healthcare GmbH
MAGNETOM Viato.Mobile (1.5T) with syngo MR XA51A	K240608, cleared on March 29, 2024	LNH, LNI, MOS	Siemens Healthcare GmbH
syngo.via VB40A	K191040, cleared on May 16, 2019	LLZ	Siemens Healthcare GmbH
Reference Device	**FDA Clearance Number and Date**	**Product Code**	**Manufacturer**
MAGNETOM Cima.X with syngo MR XA61A	K231587, cleared on December 18, 2023	LNH, LNI, MOS	Siemens Healthcare GmbH
MAGNETOM Sola with syngo MR XA61A	K232535, cleared on December 22, 2023	LNH, LNI, MOS	Siemens Healthcare GmbH
MAGNETOM Vida with syngo MR XA50A,	K213693, cleared Feb 25,2022	LNH, LNI, MOS	Siemens Healthcare GmbH
MAGNETOM Aera with syngo MR VE11	K153343, cleared on April 15, 2016	LNH, LNI, MOS	Siemens Healthcare GmbH

7. Technological Characteristics

The subject devices, MAGNETOM Avanto Fit, MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile, all with the new software syngo MR XA70A, are substantially equivalent to the predicate devices with regard to the operational environment, programming language, operating system and performance.

The subject devices conform to the standard for medical device software (IEC 62304) and other relevant IEC and NEMA standards.

There are some differences in technological characteristics between the subject devices and predicate devices, including new and modified hardware/software. Here is summary of differences:

Summary hardware comparison table for the subject and predicate device

Hardware	**Subject Devices** MAGNETOM Avanto Fit, MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile, all with software syngo MR XA70A	**Predicate Devices** MAGNETOM Avanto Fit with syngo MR XA50A (K220151), MAGNETOM Skyra Fit with syngo MR XA50A (K220589), MAGNETOM Sola Fit with syngo MR XA51A (K221733), and MAGNETOM Viato.Mobile with syngo MR XA51A (K240608)	**Reference Devices** MAGNETOM Cima.X (K231587) with syngo MR XA61A; MAGNETOM Sola (K232535) with syngo MR XA61A; MAGNETOM Vida (K213693) with syngo MR XA50A, (K203443) MAGNETOM Aera with syngo MR VE11
Magnet System	Yes	Yes	Yes
RF System	Yes	Yes	Yes
Transmission technique	Yes	Yes	Yes
Gradient System	Yes	Yes	Yes
Patient Table	Yes	Yes	Yes

[2] The predicate system MAGNETOM Sola Fit (1.5T) with syngo MR XA51A (K221733) is used in this submission as well as reference system in the Fact Sheets for some systems. However, it will not be listed again as reference system, since it is already incorporated into the predicate system overview.

Multi-Nuclear Option - Supported Nuclei	No	No	No (Yes for MAGNETOM Cima.X)
Computer	Yes	Yes	Yes
Coils	Yes New for MAGNETOM Avanto Fit, based on predicate: BM Head/Neck 20	Yes	Yes
Other HW components	Yes 3D camera, cushions modified compared to the respective subject device (see **Device Description**)	Yes	Yes

Summary software comparison table for the subject and predicate devices

Software	**Subject Devices** MAGNETOM Avanto Fit, MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile, all with software syngo MR XA70A	**Predicate Devices** MAGNETOM Avanto Fit with syngo MR XA50A (K220151), MAGNETOM Skyra Fit with syngo MR XA50A (K220589), MAGNETOM Sola Fit with syngo MR XA51A (K221733), and MAGNETOM Viato.Mobile with syngo MR XA51A (K240608)
Sequences		
SE-based pulse sequence types	New feature as listed in the **Device Description** above	Yes
GRE-based/Steady-State pulse sequence types	New or modified pulse sequences as listed in the **Device Description** above	Yes
EPI-based pulse sequence types	New features as listed in the **Device Description** above	Yes
Spectroscopy pulse sequence types	Yes	Yes
Feature and Applications		
Other features and applications such as: -Application Suites -myExam Assists -Other Imaging Applications	Modified features and applications as listed in the **Device Description** above	Yes
User interface and user interaction	Yes	Yes
Viewing and post-processing	New or modified viewing and post-processing features as listed in the **Device Description** above	Yes
Workflow and software utilization	Yes	Yes
Patient Management	Yes	Yes
Scan Modes and Pulse Sequences	Yes	Yes
Scanning	Modified and new features and applications as listed in the **Cover letter, Device Description** and **Substantial Equivalence Comparison Tables**	Yes
Reconstruction	New feature	Yes

	as listed in the **Device Description** above	
Image Display	Yes	Yes
File/Data Management	Yes	Yes

The differences have been tested and the conclusion from the non-clinical data suggests that the features bear an equivalent safety and performance profile to that of the predicate device.

8. Nonclinical Tests

The following performance testing was conducted on the subject devices:

Performance Test	Tested Hardware or Software	Source/Rationale for test
Software verification and validation	New or modified software features	Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices
Sample clinical images	New or modified software features	Guidance for submission of Premarket Notifications for Magnetic Resonance Diagnostic Devices
Image quality assessment by sample clinical images	- new / modified pulse sequence types. - comparison images between the new / modified features and the predicate device features	
myExam 3D Camera Validation and Verification Report	myExam 3D Camera	Hardware

The following performance testing for local coils was conducted on the predicate and the reference devices and can be reused for the subject device:

Performance Test	Tested Hardware or Software	Source/Rationale for test
Performance bench test	- SNR and image uniformity measurements for coils - Heating measurements for coils	Guidance for Submission of Premarket Notifications for Magnetic Resonance Diagnostic Devices

The measurements for the BM Head/Neck 20, which is added with this submission for MAGNETOM Avanto Fit were performed on an older version of the reference device and are also valid for the subject device according to the expert statements.

Overall, the results from each set of tests demonstrate that the devices perform as intended and are thus substantially equivalent to the predicate device to which it has been compared.

AI Features/Applications training and validation:

The information below shows an executive summary of training and validation dataset of the AI features:

	Deep Resolve Boost:	Deep Resolve Sharp:
Training and Validation data	• TSE: more than 25,000 slices • HASTE: pre-trained on the TSE dataset and refined with more than 10,000 HASTE slices • EPI Diffusion: more than 1,000,000 slices The data covered a broad range of body parts, contrasts, fat suppression techniques, orientations, and field strength.	on more than 10,000 high resolution 2D images. The data covered a broad range of body parts, contrasts, fat suppression techniques, orientations, and field strength.
Test Statistics and Test Results Summary	The impact of the network has been characterized by several quality metrics such as peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). Most importantly, the performance was evaluated by visual comparisons to evaluate e.g., aliasing artifacts, image sharpness and denoising levels.	The impact of the network has been characterized by several quality metrics such as peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and perceptual loss. In addition, the feature has been verified and validated by inhouse tests. These tests include visual rating and an evaluation of image sharpness by intensity profile comparisons of reconstructions with and without Deep Resolve Sharp.
Equipment	1.5T and 3T MRI systems	
Clinical Subgroups	No clinical subgroups have been defined for the collected dataset.	
Demographic Distribution	Due to reasons of data privacy, we did not record gender, age and ethnicity during data collection.	
Reference Standard	The acquired datasets (as described above) represent the ground truth for the training and validation. Input data was retrospectively created from the ground truth by data manipulation and augmentation. This process includes further under-sampling of the data by discarding k-space lines, lowering of the SNR level by addition Restricted of noise and mirroring of k-space data.	The acquired datasets represent the ground truth for the training and validation. Input data was retrospectively created from the ground truth by data manipulation. k-space data has been cropped such that only the center part of the data was used as input. With this method corresponding low-resolution data as input and high-resolution data as output / ground truth were created for training and validation.

9. Clinical Tests / Publications

No clinical tests were conducted to support substantial equivalence for the subject devices; however, as stated above, sample clinical images were provided.

Furthermore, additional clinical publications were referenced to provide information on the use of the following features and functions:

Feature	Publications
GRE_PC	[13_1] Christian Guenthner, Sweta Sethi, Marian Troelstra, Ristretto MRE: A generalized multi-shot GRE-MRE sequence NMR Biomed 2019 32:e4049
3D Whole Heart	[13_2] Henningsson M, Koken P, Stehning C et al.: Whole-heart coronary MR angiography with 2D self-navigated image reconstruction Magn Reson Med, 2012, 67:437–45 [13_3] Prieto C, Doneva M, Usman M et al.: Highly efficient respiratory motion compensated free-breathing coronary MRA using golden-step Cartesian acquisition, J Magn Reson Imaging, 2015 41:738–46 [13_4] Correia T, Ginami G, Cruz G, et al.: Optimized respiratory-resolved motion compensated 3D Cartesian coronary MR angiography, Magn Reson Med, 2018 80:2618–2629 [13_5] Chefd'hotel C, Hermosillo G, Faugeras O Flows of diffeomorphisms for multimodal image registration. Proc. IEEE International Symposium on Biomedical Imaging 2002 753–756 [13_6] Batchelor PG, Atkinson D, Irarrazaval P, et al.: Matrix Description of General Motion Correction Applied to Multishot Images, Magn Reson Med 2005 54:1273–1280 [13_7] Zeilinger MG, Kunze K-P, Munoz C et al.: Non-rigid motion-corrected free-breathing 3D myocardial Dixon LGE imaging in a clinical setting, Eur Radiol 2022 32:4340-4351
SPACE Improvement: high bandwidth IR pulse	[13_8] Wang X, Greer JS, Dimitrov IE, Pezeshk P, Chhabra A, Madhuranthakam AJ: Frequency Offset Corrected Inversion Pulse for B0 and B1 Insensitive Fat Suppression at 3T: Application to MR Neurography of Brachial Plexus, J Magn Reson Imaging, 2018 Oct 48(4):1104-1111. doi: 10.1002/jmri.26021. Epub 2018 Sep 15. PMID: 30218576
SPACE Improvement: increase gradient spoiling	[13_9] Mugler JP 3rd: Optimized three-dimensional fast-spin-echo MRI, J Magn Reson Imaging, 2014 Apr 39(4):745-67. doi: 10.1002/jmri.24542. Epub 2014 Jan 8. PMID: 24399498.
Deep Resolve Gain	[13_10] Kellman P. et alImage Reconstruction in SNR Units: A General Method for SNR Measurement MRM, 2005 54:1439 [13_11] Blu T. et al: The SURE-LET approach to image denoising, IEEE Transactions on Image Processing, 2007 6(11):2778-86
HASTE diffusion	[13_12] Ilıca AT, Hıdır Y, Bulakbaşı N, Satar B, Güvenç I, Arslan HH, Imre N. HASTE diffusionweighted MRI for the reliable detection of cholesteatoma. Diagn Interv

	Radiol. 2012 Mar-Apr 18(2):153 -8. doi: 10.4261/1 305-3825.DIR. 4246- 11.3. Epub 2011 Sep 29. PMID: 21960134
AutoMate Cardiac	[13_13] J. a. Y. S. S. a. S. M. Wetzl, "AI-based Single-Click Cardiac MRI Exam: Initial Clinical Experience and Evaluation in 44 Patients," in ISMRM & ISMRT Annual Meeting & Exhibition, 2023. [13_14] S. S. a. P. E. a. S. M. a. F. C. a. C. T. a. S. P. a. F. J. L. a. T. C. a. W. J. a. M. A. Yoon, "Automated Cardiac Resting Phase Detection Targeted on the Right Coronary Artery," Machine Learning for Biomedical Imaging, vol. 2, pp. 1-26, 2023. [13_15] S. S. a. S. M. a. R. M. a. C. T. a. S. P. a. E. T. a. T. C. a. W. R. Yoon, "Validation of a deep learning based automated myocardial inversion time selection for late gadolinium enhancement imaging in a prospective study," in ISMRM & ISMRT Annual Meeting & Exhibition, 2021. [13_16] R. a. K. T. a. S. Y. a. Y. Y. a. S. Y. S. a. W. J. a. S. M. a. K. T. Ogawa, "Neural network--based fully automated cardiac resting phase detection algorithm compared with manual detection in patients," Acta Radiologica Open, vol. 11, no. 10, p. 20584601221137772, 2022. [13_17] S. a. W. J. a. S. M. a. B. M. a. Y. S. S. a. G. C. a. B. R. McDermott, "AIbased Cardiac Scan Automation: A Prospective Comparison of Highly Automated Scan Workflows in 32 Patients," in Society for Cardiovascular Magnetic Resonance, 2024. [13_18] S. a. A. M. a. J. A. a. S. R. B. a. Z. T. a. K. M. a. S. J. a. T. E. a. C. E. a. S. C. Bohnen, "Cardiovascular magnetic resonance imaging in the prospective, population-based, Hamburg City Health cohort study: objectives and design," Journal of Cardiovascular Magnetic Resonance, vol. 20, pp. 1-11, 2018. [13_19] T. a. G. P. a. H. T. a. U. T. a. C. S. a. K. M. a. T. S. a. L. Y. a. M. M. C. a. S. F. Pezel, "Vasodilatation stress cardiovascular magnetic resonance imaging: Feasibility, workflow and safety in a large prospective registry of more than 35,000 patients," Archives of Cardiovascular Diseases, vol. 114, pp. 490-503, 2021. [13_20] G. a. P. A. U. a. K. K. P. a. N. R. a. H. R. a. W. J. a. Y. S. S. a. S. M. a. N. B. L. a. P. C. a. o. Wood, "Automated detection of cardiac rest period for trigger delay calculation for image-based navigator coronary magnetic resonance angiography," Journal of Cardiovascular
Ghost reduction	[13_21] W Scott Hoge, Jonathan R Polimeni: Dual-polarity GRAPPA for simultaneous reconstruction and

	ghost correction of echo planar imaging data Magnetic Resonance in Medicine 2016 July 76(1):32- 44

10. Safety and Effectiveness

The device labeling contains instructions for use and any necessary cautions and warnings to ensure safe and effective use of the device.

Risk Management is ensured via a risk analysis in compliance with ISO 14971, to identify and provide mitigation of potential hazards early in the design cycle and continuously throughout the development of the product. Siemens Healthcare GmbH adheres to recognized and established industry standards, such as the IEC 60601-1 series, to minimize electrical and mechanical hazards. Furthermore, the device is intended for healthcare professionals familiar with and responsible for the acquisition and post processing of magnetic resonance images.

The subject devices conform to the following FDA recognized and international IEC, ISO and NEMA standards:

Recognition Number	Product Area	Title of Standard	Reference Number and date	Standards Development Organization
19-4	General	Medical electrical equipment - part 1: general requirements for basic safety and essential performance	ES60601-1:2005/(R)2012 and A1:2012 C1:2009/(R)2012	AAMI / ANSI
19-8	General	Medical electrical equipment - Part 1-2: General requirements for basic safety and essential performance - Collateral Standard: Electromagnetic disturbances - Requirements and tests	60601-1-2 Edition 4.0:2014-02	IEC
12-295	Radiology	Medical electrical equipment - Part 2-33: Particular requirements for the basic safety and essential performance of magnetic resonance equipment for medical diagnosis	60601-2-33 Ed. 3.2 b:2015	IEC
5-125	General	Medical devices - Application of risk management to medical devices	14971 Third Edition 2019-12	ISO
5-114	General I (QS/ RM)	Medical devices - Part 1: Application of usability engineering to medical devices	62366-1:2015	ANSI AAMI IEC
13-79	Software/ Informatics	Medical device software - Software life cycle processes	62304 Edition 1.1 2015-06 CONSOLIDATED VERSION	IEC
12-195	Radiology	NEMA MS 6-2008 (R2014) Determination of Signal-to-Noise Ratio and Image Uniformity for Single-Channel Non-Volume Coils in Diagnostic MR Imaging	MS 6-2008 (R2014)	NEMA
12-349	Radiology	Digital Imaging and Communications in Medicine (DICOM)	PS 3.1 - 3.20 (2022d)	NEMA

2-258	Biocompatibility	Biological evaluation of medical devices - part 1: evaluation and testing within a risk management process. (Biocompatibility)	10993-1 Fifth edition 2018-08	AAMI ANSI ISO

11. Conclusion as to Substantial Equivalence

MAGNETOM Avanto Fit, MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile all with software syngo MR XA70A have the same intended use and same basic technological characteristics than the respective predicate device system, MAGNETOM Avanto Fit with syngo MR XA50A, MAGNETOM Skyra Fit with syngo MR XA50A, MAGNETOM Sola Fit with syngo MR XA51, and MAGNETOM Viato.Mobile with syngo MR XA51A, with respect to the magnetic resonance features and functionalities. While there are some differences in technical features compared to the predicate devices, the differences have been tested and the conclusions from all verification and validation data suggest that the features bear an equivalent safety and performance profile to that of the predicate device and reference devices.

Siemens believes that the subject devices (MAGNETOM Avanto Fit , MAGNETOM Skyra Fit, MAGNETOM Sola Fit, and MAGNETOM Viato.Mobile all with software syngo MR XA70A) are substantially equivalent to the currently marketed devices MAGNETOM Avanto Fit with syngo MR XA50A (K220151, cleared on April 01, 2022), MAGNETOM Skyra Fit with syngo MR XA50A (K220589, cleared on May 13, 2022), MAGNETOM Sola Fit with syngo MR XA51A (K221733, cleared on September 13, 2022), and MAGNETOM Viato.Mobile with syngo MR XA51A (K240608, cleared on March 29, 2024).